Exercises 7.1

39. a. Show that $f(x) = x^3$ and $g(x) = \sqrt[3]{x}$ are inverses of one another.

b. Graph f and g over an x-interval large enough to show the graphs intersecting at $(1, 1)$ and $(-1, -1)$. Be sure the picture shows the required symmetry about the line $y = x$.

c. Find the slopes of the tangents to the graphs of f and g at $(1, 1)$ and $(-1, -1)$ (four tangents in all).

d. What lines are tangent to the curves at the origin?

59. Assume that f and g are differentiable functions that are inverses of one another so that $(g \circ f)(x) = x$. Differentiate both sides of this equation with respect to x using the Chain Rule to express $(g \circ f)'(x)$ as a product of derivatives of g and f. What do you find? (This is not a proof of Theorem 1 because we assume here the theorem's conclusion that $g = f^{-1}$ is differentiable.)

60. Equivalence of the washer and shell methods for finding volume Let f be differentiable and increasing on the interval $a \le x \le b$, with $a > 0$, and suppose that f has a differentiable inverse, f^{-1}. Revolve about the y-axis the region bounded by the graph of f and the lines $x = a$ and $y = f(b)$ to generate a solid. Then the values of the integrals given by the washer and shell methods for the volume have identical values:

$$\int_{f(a)}^{f(b)} \pi((f^{-1}(y))^2 - a^2)\, dy = \int_a^b 2\pi x(f(b) - f(x))\, dx.$$

To prove this equality, define

$$W(t) = \int_{f(a)}^{f(t)} \pi((f^{-1}(y))^2 - a^2)\, dy$$

$$S(t) = \int_a^t 2\pi x(f(t) - f(x))\, dx.$$

Then show that the functions W and S agree at a point of $[a, b]$ and have identical derivatives on $[a, b]$. As you saw in Section 4.7, Exercise 90, this will guarantee $W(t) = S(t)$ for all t in $[a, b]$. In particular, $W(b) = S(b)$. (*Source:* "Disks and Shells Revisited," by Walter Carlip, *American Mathematical Monthly*, Vol. 98, No. 2, Feb. 1991, pp. 154–156.)

Exercises 7.2

Finding Derivatives

In Exercises 5–36, find the derivative of y with respect to x, t, or θ, as appropriate.

36. $y = \displaystyle\int_{\sqrt{x}}^{\sqrt[3]{x}} \ln t\, dt$

Evaluating Integrals

Evaluate the integrals in Exercises 37–54.

54. $\displaystyle\int \frac{\sec x\, dx}{\sqrt{\ln(\sec x + \tan x)}}$

70. **a.** Prove that $f(x) = x - \ln x$ is increasing for $x > 1$.

b. Using part (a), show that $\ln x < x$ if $x > 1$.

73. The region in the first quadrant bounded by the coordinate axes, the line $y = 3$, and the curve $x = 2/\sqrt{y+1}$ is revolved about the y-axis to generate a solid. Find the volume of the solid.

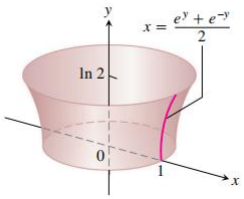
Exercises 7.3

Evaluate the integrals in Exercises 93–106.

104. $\displaystyle\int_2^3 \frac{2\log_2(x-1)}{x-1}\,dx$

123. Find the absolute maximum value of $f(x) = x^2 \ln(1/x)$ and say where it is assumed.

128. Find the area of the surface generated by revolving the curve $x = (e^y + e^{-y})/2, 0 \le y \le \ln 2$, about the y-axis.

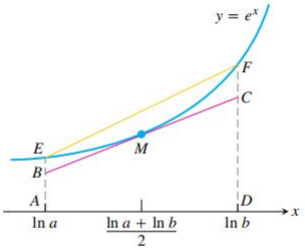


136. The geometric, logarithmic, and arithmetic mean inequality

a. Show that the graph of e^x is concave up over every interval of x-values.

b. Show, by reference to the accompanying figure, that if $0 < a < b$, then

$$e^{(\ln a + \ln b)/2} \cdot (\ln b - \ln a) < \int_{\ln a}^{\ln b} e^x \, dx < \frac{e^{\ln a} + e^{\ln b}}{2} \cdot (\ln b - \ln a).$$



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c. Use the inequality in part (b) to conclude that

$$\sqrt{ab} < \frac{b - a}{\ln b - \ln a} < \frac{a + b}{2}.$$

This inequality says that the geometric mean of two positive numbers is less than their logarithmic mean, which in turn is less than their arithmetic mean.

Exercises 8.2

Using Substitution

Evaluate the integrals in Exercise 25–30 by using a substitution prior to integration by parts.

28. $\int \ln (x + x^2)\, dx$

Evaluating Integrals

Evaluate the integrals in Exercises 31–52. Some integrals do not require integration by parts.

50. $\int_0^{1/\sqrt{2}} 2x \sin^{-1} (x^2)\, dx$

Reduction Formulas

In Exercises 63–67, use integration by parts to establish the reduction formula.

67. $\int x^m (\ln x)^n\, dx = \frac{x^{m+1}}{m+1} (\ln x)^n - \frac{n}{m+1} \cdot$

$\int x^m (\ln x)^{n-1}\, dx, \quad m \neq -1$

Exercises 75 and 76 compare the results of using Equations (4) and (5).

75. Equations (4) and (5) give different formulas for the integral of $\cos^{-1} x$:

a. $\int \cos^{-1} x\, dx = x \cos^{-1} x - \sin (\cos^{-1} x) + C$ Eq. (4)

b. $\int \cos^{-1} x\, dx = x \cos^{-1} x - \sqrt{1 - x^2} + C$ Eq. (5)

Can both integrations be correct? Explain.

Evaluate the integrals in Exercises 77 and 78 with (**a**) Eq. (4) and (**b**) Eq. (5). In each case, check your work by differentiating your answer with respect to x.

77. $\int \sinh^{-1} x\, dx$

Exercises 7.6

Evaluate the integrals in Exercises 67–80.

80. $\int \frac{dx}{(x-2)\sqrt{x^2-4x+3}}$

Verify the integration formulas in Exercises 99–102.

99. $\int \frac{\tan^{-1} x}{x^2}\, dx = \ln x - \frac{1}{2}\ln(1+x^2) - \frac{\tan^{-1} x}{x} + C$

101. $\int (\sin^{-1} x)^2\, dx = x(\sin^{-1} x)^2 - 2x + 2\sqrt{1-x^2}\sin^{-1} x + C$

Find the volumes of the solids in Exercises 123 and 124.

123. The solid lies between planes perpendicular to the x-axis at $x = -1$ and $x = 1$. The cross-sections perpendicular to the x-axis are

a. circles whose diameters stretch from the curve $y = -1/\sqrt{1 + x^2}$ to the curve $y = 1/\sqrt{1 + x^2}$.

b. vertical squares whose base edges run from the curve $y = -1/\sqrt{1 + x^2}$ to the curve $y = 1/\sqrt{1 + x^2}$.

Exercises 7.7

11. Prove the identities

$$\sinh(x + y) = \sinh x \cosh y + \cosh x \sinh y,$$
$$\cosh(x + y) = \cosh x \cosh y + \sinh x \sinh y.$$

Then use them to show that

a. $\sinh 2x = 2 \sinh x \cosh x$.

b. $\cosh 2x = \cosh^2 x + \sinh^2 x$.

Verify the integration formulas in Exercises 37–40.

37. a. $\int \operatorname{sech} x\, dx = \tan^{-1}(\sinh x) + C$

b. $\int \operatorname{sech} x\, dx = \sin^{-1}(\tanh x) + C$

40. $\int \tanh^{-1} x\, dx = x \tanh^{-1} x + \frac{1}{2}\ln(1 - x^2) + C$

76. Derive the formula $\sinh^{-1} x = \ln\left(x + \sqrt{x^2 + 1}\right)$ for all real x. Explain in your derivation why the plus sign is used with the square root instead of the minus sign.